# Loan Default Prediction

An interpretable credit-scoring model to approve home equity loans

Capstone Project | Practical Data Science | MIT Professional Education - Applied AI and Data Science Program

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# Key takeaways

**1 in 5**

home equity loans in the data ended in default. Accuracy alone is a misleading yardstick.

**62%**

default rate when the applicant's debt-to-income ratio is missing, versus 8.6% when it is present. Unverified income is the strongest warning sign.

**93%**

of future defaulters among unseen applicants. The proposed Random Forest model catches them. 2 of every 3 applicants it flagged really defaulted.

**~$9M**

indicative annual benefit on 6,000 applications, from roughly $10.8M of default-related cost down to about $1.8M.

**Recommendation:** make income and valuation checks mandatory. Deploy the forest at a 0.15 threshold with approve / refer / decline bands, after a two-month shadow run and a fairness review.

# The problem: approving home equity loans without losing the profit to defaults

- Interest on home loans is a major share of retail bank profit. Defaults consume it.
- Today each decision is manual: slow, inconsistent across underwriters, open to error and bias.
- The consumer credit department wants an empirically derived credit-scoring model built on its own recent loans.
- Constraint: under the Equal Credit Opportunity Act every rejection needs a specific, defensible reason, so the model must be interpretable.

*Objective: predict which applicants are likely to default. Tell the bank which attributes to weigh when approving a loan.*

**The data: Home Equity dataset (HMEQ)**

**5,960**
recent loans

**12**
attributes known at application

**19.9%**
defaulted or seriously delinquent

**44%**
of applications have a blank field

# Approach: from raw applications to a decision the bank can defend

1

## Explore

Univariate, bivariate and correlation analysis. Default rate by band. Pattern of missing values

2

## Prepare

Reserve a stratified 30% test set first. Flag every missing field. Impute median/mode learned on training rows. Cap outliers for the linear model only. One-hot encode

3

## Model

Logistic Regression (linear benchmark), Decision Tree, Bagging, Random Forest. Class weights. Cross-validated grid search

4

## Measure

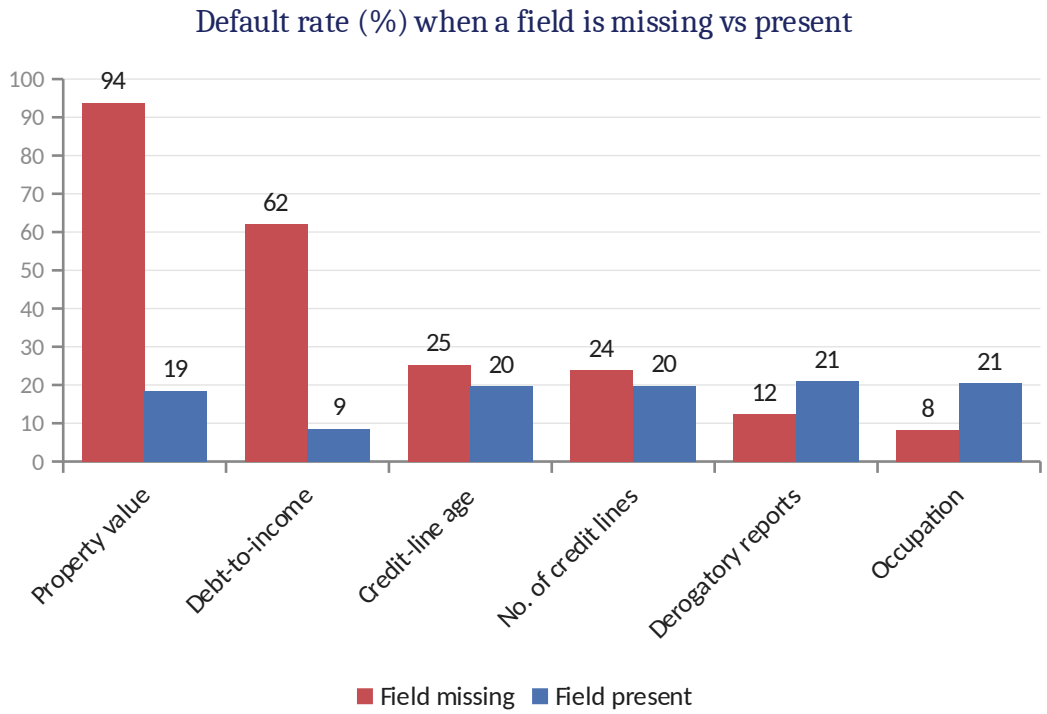
Recall on defaulters as the primary metric. Precision, F1, ROC-AUC as guard-rails. Final model and threshold chosen on out-of-fold expected cost

5

## Recommend

Model specification, business rules, implementation plan, benefits, risks

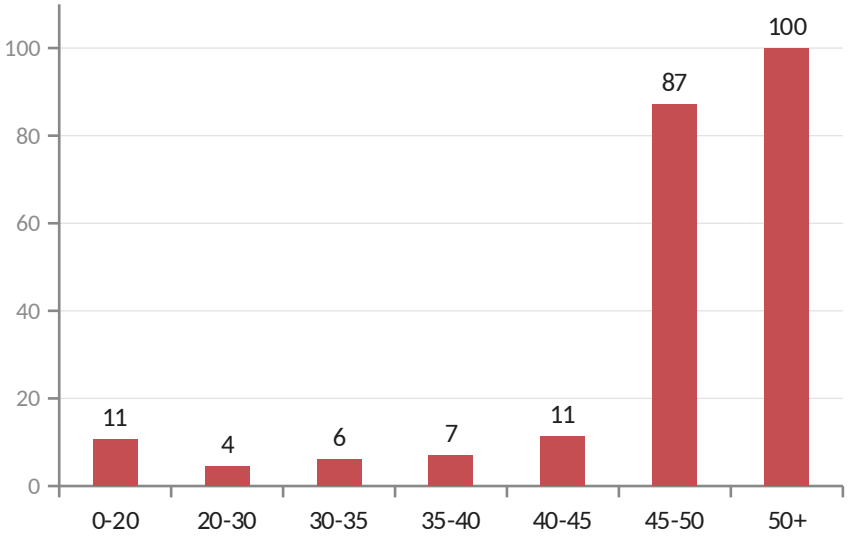
# Insight 1: information the applicant did not provide is the strongest warning sign



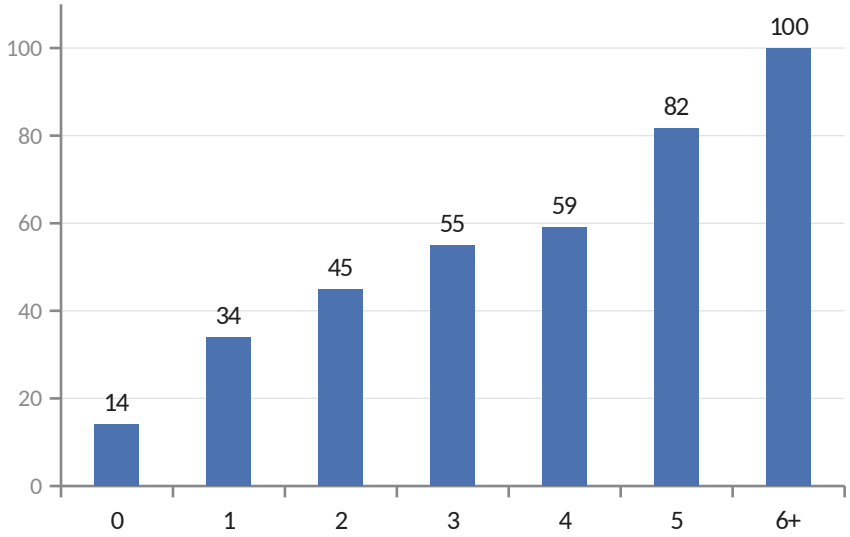

- No recorded debt-to-income ratio: 62% default (1,267 applicants, 21% of the data).
- No property valuation: 94% default (112 applicants).
- The overall default rate is 20%, so these gaps multiply risk 3 to 5 times.
- Interpretation: the bank could not check the income or collateral, or the applicant chose not to supply it.
- Action: every imputed field gets a 'was missing' flag in the model. Verification should become mandatory in the process.

# Insight 2: debt burden has a cliff at 45%, and past delinquency repeats

Default rate (%) by debt-to-income ratio band (%)

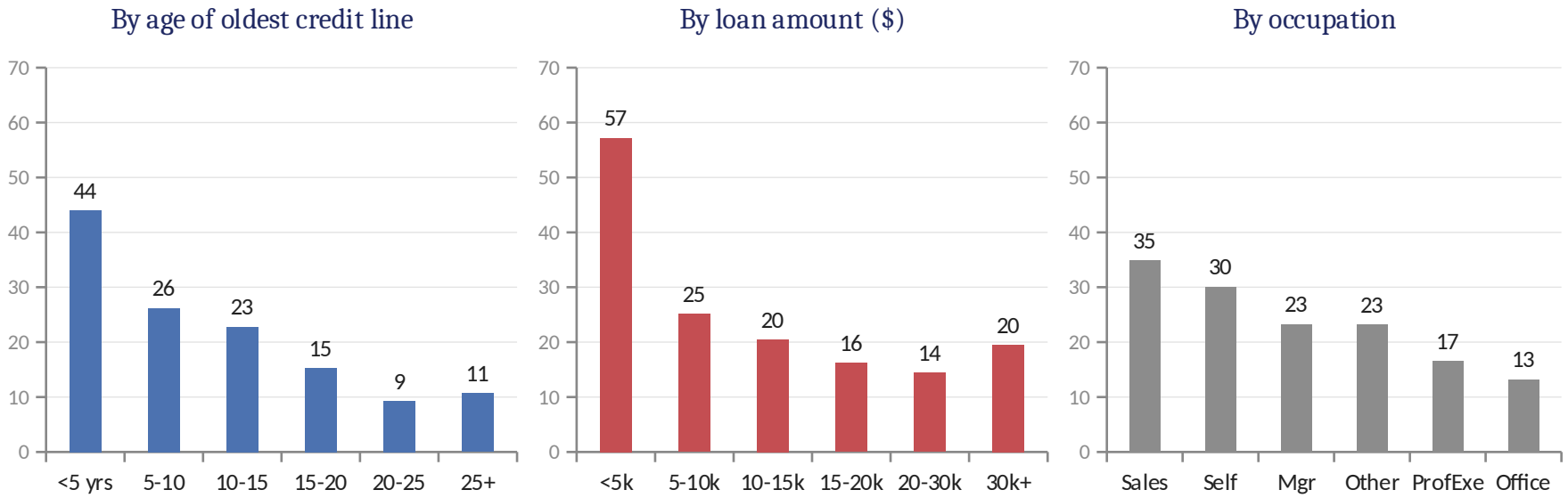



Default rate (%) by delinquent credit lines




Between a 20% and a 40% ratio only 4-7% default. Above 45% almost everyone does. The first delinquent line or derogatory report more than doubles the risk. Each further one raises it again (delinquent lines: 14% at zero, 34% at one, 100% at six or more).

# Insight 3: long credit histories protect, small loans do not, occupation matters less




Default rate (%) by group. Applicants with under five years of credit history default four times as often as those with twenty-plus years. Loans under $5,000 default 57% of the time. Occupation shifts risk moderately and adds little on its own at the usual threshold. But removing it from the final model raises the expected cost of credit errors by about 10%. So it stays, subject to the fair-lending review.

# Measure of success: the two mistakes do not cost the same

### Missed defaulter

The model says 'will repay'. The customer defaults.

**The bank loses most of the principal plus collection cost: about $9,000 on an average $16,900 defaulted loan.**

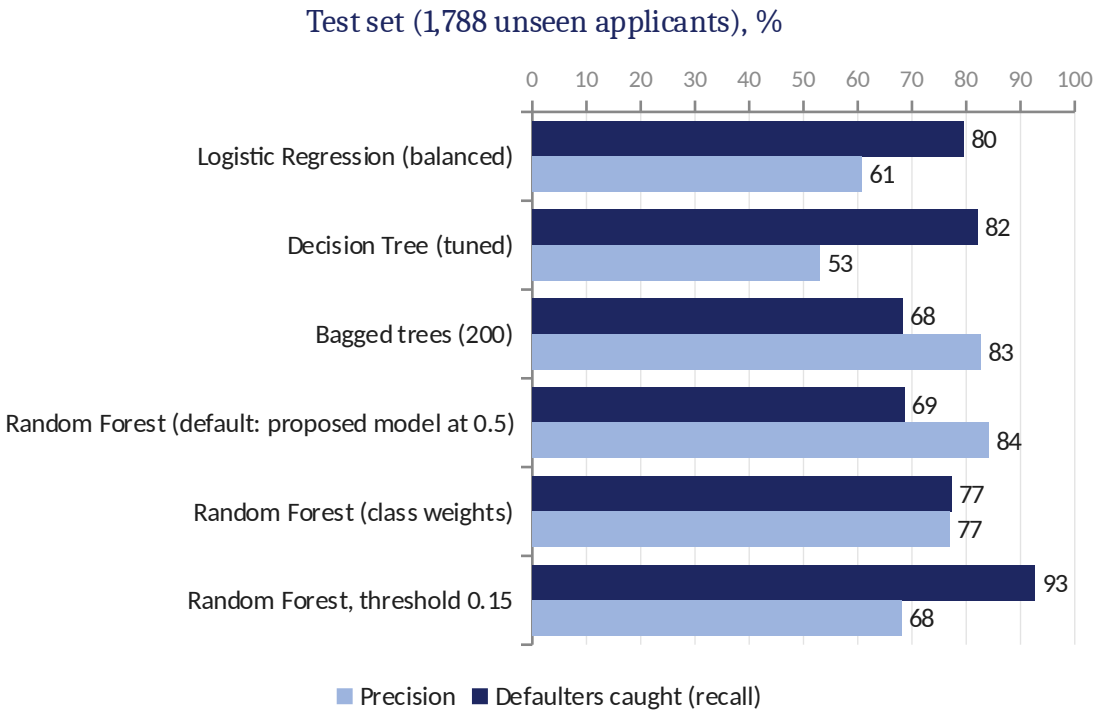
### Wrongly declined

The model says 'will default'. The customer is creditworthy.

**The bank forgoes the interest margin on one good applicant: about $2,000.**

- A missed defaulter costs roughly 4.5 times a wrongly declined customer. So the primary metric is Recall: the share of future defaulters the model catches.
- Guard-rails: Precision (how many flagged applicants really default) and F1, so the model does not simply decline everyone. ROC-AUC measures how well the model ranks applicants by risk, which matters when the threshold moves.
- We tune individual models by cross-validated grid search on recall. We choose the final model and its threshold on out-of-fold expected cost, training rows only. We score the 30% test set once.

# Model comparison: the Random Forest gives the best balance




- Linear vs non-linear: logistic regression reaches 80% recall but only 61% precision. The data is made of steps, which trees capture and a straight line cannot.
- Single tree vs ensembles: the tuned tree catches 82% but declines almost one good customer per defaulter caught (53% precision). Bagging 200 trees lifts precision to 83% and AUC to 0.95 at 68% recall.
- The class-weighted forest: 77% recall at 77% precision, ROC-AUC 0.96, among the best at ranking applicants by risk.
- Final choice on the expected cost of mistakes (training rows only), by a rule fixed beforehand: lowest mean cost, simplest if within noise. Three forests tied. The plain forest at 0.15 catches 93% at 68% precision on test.

# Proposed solution: a Random Forest scorer explained by a surrogate decision tree

## Model specification

- Scoring engine: Random Forest, 200 fully grown trees, no class weights (the threshold handles the imbalance).
- Inputs: all 27 (10 attributes, 11 missing-value flags, occupation and loan purpose). Fallback without occupation: 21 inputs, about 10% more cost.
- Threshold 0.15, chosen with the configuration on repeated out-of-fold expected cost (training rows). It holds for cost ratios 3:1 to 6:1 and across splits. Approve / refer / decline bands (69% / 7% / 24%) keep an underwriter in the loop.
- Explanation layer: a depth-4 surrogate tree reproduces 93% of the forest's out-of-fold decisions and gives the plain-language summary. Per-applicant risk drivers (built before the shadow run) give each decline its ECOA reason.
- We scored the reserved test set once, after all choices.

**93%**
of defaulters caught

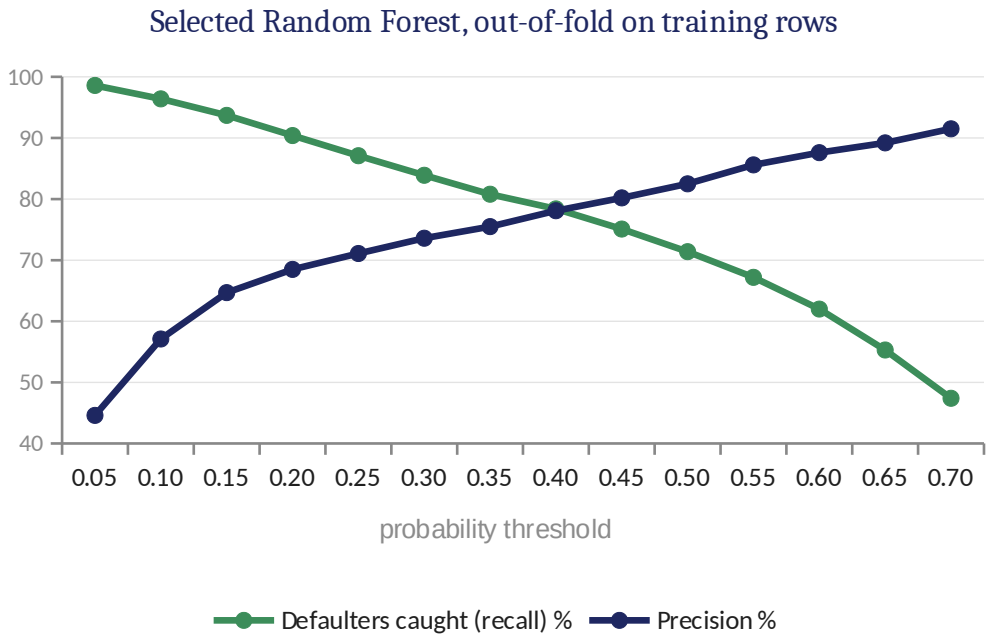
**68%**
of flagged applicants truly default

**0.96**
ROC-AUC: risk ranking quality

**90%**
overall accuracy

*Per 1,000 applicants at 0.15: 185 of 200 defaulters flagged in advance, 15 missed, 87 good customers flagged. 27% of applicants flagged, 73% approved.*

# Why a 0.15 threshold: it is a business dial, not a constant




### At 0.50 (default)

- 63 missed defaulters
- 26 good customers flagged
- $615k indicative cost

### At 0.15 (proposed)

- 15 missed defaulters
- 87 good customers flagged
- $304k indicative cost

Curves: out-of-fold predictions on the training rows. Cards: test set, per 1,000 applicants, at $9,000 per missed defaulter and $2,000 per wrongly declined customer. The optimum stays at 0.15 for cost ratios 3:1 to 6:1 and on other splits.

# Business impact: an indicative $9 million a year

| Per year, 6,000 applications | Current process | With the proposed model |
|---|---|---|
| Loans that default | about 1,200 (20%) | about 90 missed |
| Default-related losses | about $10.8M | about $0.8M |
| Good customers flagged at 0.15 | none | about 520, margin forgone about $1.0M |
| **Total cost of credit errors** | **about $10.8M** | **about $1.8M** |

**~$9M**

indicative gross annual benefit. About $8.9M after running costs, $8.5M in year one

- Assumptions: 6,000 applications a year. Defaulted loans average $16,900 with about half recovered ($9,000 per missed defaulter). Forgone margin about $2,000 per declined good customer. Production performance equal to test performance. Measured as an extra screen on the loans the bank approves today, before mandatory income verification.
- Costs: roughly $300-400k one-off (pipeline, integration, validation, fairness review, training) and about $100k a year to run and monitor. It repays its cost within year one even if the bank gets only a quarter of the benefit.
- Non-financial: faster decisions, consistent treatment, auditable reasons for every decline, verified income and valuation on every application.

# Recommendations and implementation roadmap

### 1 Months 0-2: prepare

- Make income and valuation verification mandatory fields
- Build the scoring pipeline and per-applicant risk drivers
- Check the threshold with real loss and margin figures
- Review fairness and model risk

### 2 Months 2-4: shadow run

- Score every application alongside the manual decision
- Test the model on applicants with a recorded ratio. Tune the approve / refer / decline bands
- Train underwriters on risk drivers and referrals

### 3 Month 4 onwards: scale

- Start automated approve / decline for clear cases
- Monitor monthly: recall, precision, approval rate, missing fields
- Re-train once 12+ months of verified outcomes exist, then yearly. Threshold review with finance

## Policy rules to adopt now, independent of the model

- Decline or escalate above a 45% debt-to-income ratio. Refer to an underwriter with two or more delinquent credit lines or two or more major derogatory reports. Observed default: 45% and 51%+. Where a rule and the score disagree, the stricter outcome applies.
- Treat an application with no verifiable income or property valuation as high risk, never as neutral.

# Risks, challenges and further analysis

## Risks and mitigations

- Selection bias: the data covers only loans the old process approved. Mitigate with a shadow period on all applications and reject inference.
- The 'ratio missing' signal will change once verification is mandatory. Before go-live, test the model on applicants with a recorded ratio (83% caught at 80% precision). Re-train after 12+ months of verified outcomes.
- Economic drift: monitor monthly, re-train yearly, review the threshold.
- Fair lending: the 0.15 threshold wrongly flags about 11% of creditworthy applicants (3% at 0.5). Test for disparate impact before go-live, yearly and at each threshold change. A fallback without occupation is ready (about 10% more cost).
- Explainability and automation bias: per-applicant risk drivers (built in months 0-2), a referral band, documented overrides.

## Further analysis

- Reject inference to cover applicants the old process declined (needs 12+ months of data).
- Engineered features: combined loan-to-value ratio, debt-to-income bands, interaction terms.
- Gradient boosting and probability calibration, so a score of 0.15 means exactly a 15% chance of default.
- Time-based validation once application dates are available.
- Risk-based pricing for the referral band. Early-warning scoring on existing loans.
- Applicant-friendly wording for the per-applicant risk drivers, tested with compliance staff.

# Thank you

Questions and discussion

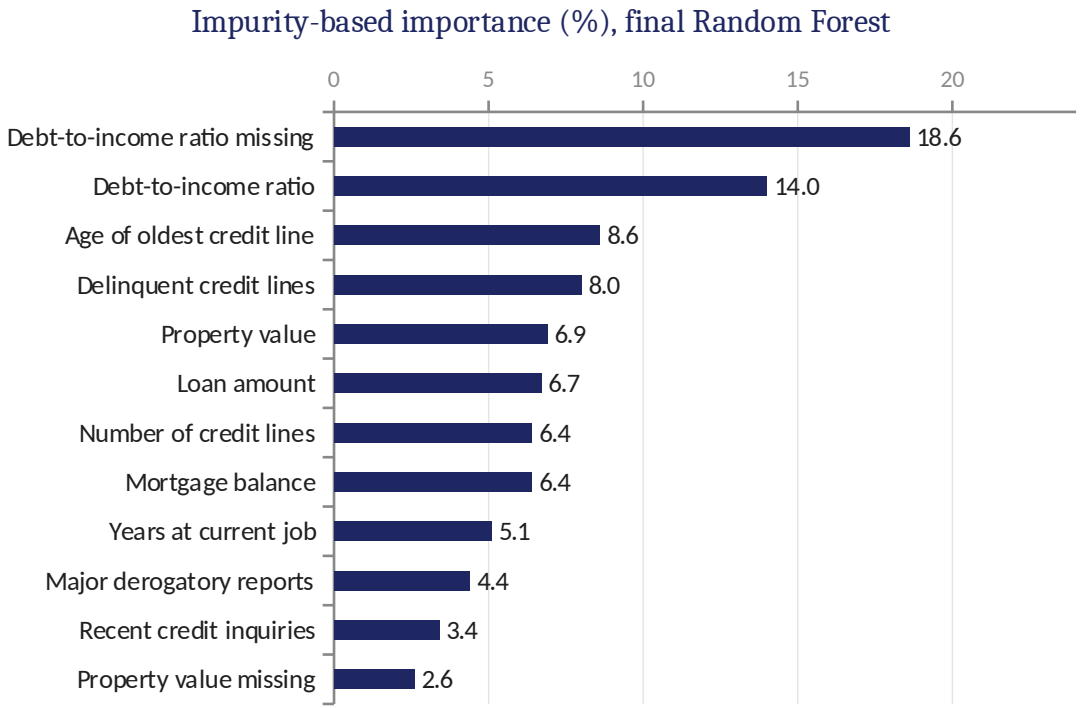
Catch 93% of defaulters. Explain every decision. Save an indicative $9M a year.

# Appendix A: all models on the held-out test set

| Model | Recall | Precision | F1 | Accuracy | ROC-AUC | Train recall |
|---|---|---|---|---|---|---|
| Logistic Regression (plain) | 0.566 | 0.774 | 0.654 | 0.880 | 0.907 | 0.635 |
| Logistic Regression (balanced, C tuned) | 0.796 | 0.608 | 0.689 | 0.857 | 0.906 | 0.821 |
| Decision Tree (default) | 0.686 | 0.706 | 0.696 | 0.880 | 0.807 | 1.000 |
| Decision Tree (class weights) | 0.616 | 0.719 | 0.664 | 0.875 | 0.778 | 1.000 |
| Decision Tree (tuned: balanced, min 20 per leaf) | 0.821 | 0.531 | 0.645 | 0.819 | 0.877 | 0.921 |
| Bagging (200 fully grown trees) | 0.683 | 0.827 | 0.748 | 0.908 | 0.945 | 1.000 |
| Bagging (class-weighted trees) | 0.597 | 0.810 | 0.687 | 0.891 | 0.946 | 1.000 |
| Random Forest (default, 200 trees) | 0.686 | 0.842 | 0.756 | 0.912 | 0.963 | 1.000 |
| Random Forest (class weights 0.2/0.8) | 0.773 | 0.769 | 0.771 | 0.908 | 0.959 | 1.000 |
| Random Forest (tuned for recall) | 0.790 | 0.656 | 0.717 | 0.875 | 0.932 | 0.918 |
| Random Forest (tuned, 12 features) | 0.784 | 0.660 | 0.717 | 0.876 | 0.929 | 0.916 |
| **Proposed: RF unweighted, 27 inputs, threshold 0.15** | **0.927** | **0.681** | **0.785** | **0.899** | **0.963** | - |

Test set of 1,788 applicants at the 0.5 threshold except the proposed model. Train recall of 1.000 marks un-pruned models that memorised the training rows. We chose the proposed model and its threshold on repeated out-of-fold expected cost (training rows only).

# Appendix B: what the Random Forest relies on




- The debt-to-income ratio and its missing flag carry a third of the final forest's decision power. With delinquent lines and credit-history length, about half.
- Permutation importance (drop in ranking quality when we shuffle a column) ranks the same four at the top, delinquent credit lines first.
- Loan, property and mortgage amounts matter in combination, little on their own.
- Occupation, loan purpose and minor flags matter little individually. Yet removing them raised out-of-fold expected cost by about a fifth, and removing occupation alone by about 10%. So they stay subject to the fair-lending review.
- Logistic-regression coefficients, tree rules and the surrogate tree all point to the same drivers.

# Appendix C: the explanation layer, rules of the surrogate tree that follows the forest

| | | |
|---|---|---|
| 1 | **Is the debt-to-income ratio missing?** | Yes: flag, whatever the credit history (62% of these applicants defaulted). Check income before approving. |
| 2 | **If recorded, is the ratio above about 44%?** | Yes: flag (87-100% of applicants above 45% defaulted). |
| 3 | **Ratio at or below 44%: a delinquent credit line together with a major derogatory report?** | Yes: flag. Otherwise clear. Applicants with no delinquent line default only about 4-5% of the time. |
| 4 | **How faithful are these rules?** | They reproduce 93% of the forest's decisions on unseen applicants (73% for never flagging, kappa 0.82). Per-applicant risk drivers from the forest give each decline its own reason. |
| 5 | **What the rules do not use** | Loan size, credit-history length, occupation and loan purpose do not change the surrogate's outcome. They matter only in combination inside the forest. |

Surrogate: depth-4 tree, at least 20 applicants per leaf, fitted to the forest's out-of-fold flag/clear decisions at threshold 0.15 on the training rows. The stand-alone tuned tree (test recall 82%, precision 53%) found the same drivers.

# Appendix D: data treatment and pre-processing

| Step | What we did | Why |
|---|---|---|
| Duplicates | None found. No rows removed | 5,960 loans retained, including 2 with only the loan amount recorded |
| Missing values | A 0/1 'was missing' flag per column, then median (numeric) or mode (categorical) imputation learned on the training rows | Missingness of debt-to-income and property value is the strongest default signal. Removing rows would lose 44% of the data |
| Outliers | IQR capping for the linear model only. DEROG and DELINQ capped at the 99th percentile. Tree models use raw values | Extreme values pull linear models (debt-to-income 203%, credit history 97 years). Trees split on rank order and keep the 'above 50% always defaults' signal |
| Encoding and scaling | One-hot encoding of REASON and JOB (drop first). Standard scaling of the continuous inputs for the linear model only, flags left as 0/1 | Comparable coefficients, odds ratios for flags read as the effect of the flag |
| Split | 70% train / 30% test, stratified on the target, fixed seed, reserved before we fitted any statistic. Identical applicants in both data versions | No leakage from test rows into fences, medians or scaling. Results comparable across models |
| Imbalance | Class weights (balanced or 0.2/0.8) inside the individual models, then the decision threshold for the final model. No resampling | Keeps each real applicant. Class weights lifted recall by 23 points for logistic regression and 9 for the forest at the 0.5 threshold. They added nothing once we chose the threshold on cost, so the final forest is unweighted |
| Model inputs | 27 columns: the 10 attributes, 11 missing flags and 6 one-hot indicators (REASON HomeImp, JOB Office/Other/ProfExe/Sales/Self) | The notebook's final-pipeline cell names every input to the final model, with the imputation values and model settings |